

Spatiotemporal spectral derivatives with forward stepwise BIC for robust partial differential equation discovery

Fuchang Wang^{a,*}, Huirong Cao^b

^a*School of Science, Emergency Management University, No. 465 College Road, Sanhe, Hebei 065201, China*

^b*School of Science, Langfang Normal University, No. 100 Ai-min Road, Langfang, Hebei 065000, China*

Abstract

Data-driven discovery of partial differential equations (PDEs) from noisy data faces two persistent challenges. The first is inaccurate derivative computation. The second is threshold-sensitive sparse regression. This paper presents a spectral weak formulation that addresses both.

Spatial and temporal derivatives are computed through a two-dimensional Fourier transform over the space-time domain. This achieves spectral accuracy. It also resolves non-periodic traveling-wave solutions, where spatial-only spectral methods fail.

The sparse regression is formulated in a weak sense. The PDE residual is integrated against localized Gaussian test functions. This averaging suppresses high-frequency noise in high-order derivatives. Candidate terms are normalized to unit ℓ_2 norm. Forward stepwise regression with the Bayesian information criterion then selects the sparse model automatically. No threshold tuning is required.

Experiments on the MDBench benchmark show clear improvements. The method recovers both true terms of the Burgers equation at all noise levels, including snr 10 where pointwise regression loses the small diffusion term. It achieves exact recovery of the KdV equation at three of five noise levels. It recovers all three terms of the Kuramoto–Sivashinsky equation at moderate noise. On the non-periodic advection equation, the method achieves exact

*Corresponding author. Tel.: +86-xxx-xxxx-xxxx; email: wangfuchang@cidp.edu.cn

recovery at clean while finite-difference baselines fail completely. The framework is 10–20 times faster than Weak SINDy with comparable accuracy.

Keywords: PDE discovery, spectral methods, sparse regression, Bayesian information criterion, spatiotemporal derivatives, symbolic regression

1. Introduction

Recovering the governing equations of physical systems from data lies at the heart of computational science. When first-principles derivation is intractable, researchers turn to data-driven methods to infer partial differential equations (PDEs) from spatio-temporal measurements. The growing volume of high-fidelity simulation and experimental data has made this approach increasingly practical.

Rudy et al. [1] introduced the sparse identification of nonlinear dynamics (SINDy) framework. It casts PDE discovery as a sparse regression problem. From data $u(\mathbf{x}, t)$, one builds a library $\Theta(u)$ of candidate terms: polynomials and spatial derivatives. One then solves $\mathbf{u}_t = \Theta(u)\boldsymbol{\xi}$ for a sparse coefficient vector $\boldsymbol{\xi}$. The sequential thresholded least squares (STLSQ) algorithm has become the standard solver. It alternates ridge regression with hard thresholding.

Three limitations persist. First, finite difference (FD) derivatives accumulate truncation errors at high order. Second, STLSQ requires careful threshold tuning and is noise-sensitive. Third, spectral methods are spectrally accurate but assume periodic boundaries. They degrade through Gibbs phenomena on non-periodic data.

Existing remedies include exponentially filtered spectral derivatives [2], the weak SINDy formulation [3], and deep learning approaches such as physics-informed neural networks (PINNs) [4] and neural operators [5]. These either incur substantial computational cost or sacrifice the interpretability of sparse regression.

We propose a robust PDE discovery framework with three contributions:

1. **Spatiotemporal spectral derivatives:** A two-dimensional Fourier transform over the space-time domain computes spatial and temporal derivatives simultaneously. Unlike spatial-only spectral methods, it handles non-periodic traveling-wave solutions accurately.
2. **Spectral weak formulation:** The sparse regression is formulated in weak form. The residual is integrated against localized Gaussian test

functions. This averaging suppresses high-frequency noise in high-order derivatives such as u_{xxxx} while retaining spectral accuracy.

3. **Normalized forward stepwise BIC:** Each candidate term is normalized to unit ℓ_2 norm before regression. Forward stepwise selection with the Bayesian information criterion (BIC) then determines sparsity automatically. No threshold tuning is required.

The paper is organized as follows. Section 2 reviews related work. Section 3 presents the methodology, including theoretical analysis in Section 3.3. Section 4 reports experimental results. Section 5 discusses findings and limitations. Section 6 concludes.

2. Related work

2.1. Sparse regression for PDE discovery

The SINDy framework [1] formulates PDE discovery as a sparse regression problem. Given a library of candidate terms $\Theta(u) \in \mathbb{R}^{N \times P}$ and the time derivative $\mathbf{u}_t \in \mathbb{R}^N$, the goal is to find a sparse coefficient vector $\boldsymbol{\xi} \in \mathbb{R}^P$ such that $\mathbf{u}_t \approx \Theta(u)\boldsymbol{\xi}$. The STLSQ algorithm solves this by iteratively: (1) solving a ridge regression, (2) thresholding small coefficients to zero, and (3) re-solving the least squares problem on the remaining support.

Several variants improve robustness. The sparse relaxed regularized regression (SR3) method [6] introduces a relaxation variable. This decouples the sparsity penalty from the data fidelity term and improves convergence. The weak SINDy formulation [3] replaces pointwise derivatives with integrals against test functions. This significantly reduces noise sensitivity. Trajectory-based methods [7] use multiple initial conditions to improve identifiability.

2.2. Spectral methods for derivative computation

Spectral methods offer exponential convergence for smooth functions. They are natural candidates for derivative computation in PDE discovery. Schaeffer and McCalla [2] used Fourier spectral derivatives with exponential filtering to compute high-order derivatives robustly. However, standard spectral methods assume periodic boundary conditions. They suffer from Gibbs phenomena for non-periodic data.

To address non-periodic problems, Chebyshev spectral methods have been proposed [8]. These require data sampled on Chebyshev–Gauss–Lobatto

grids. Experimental data rarely use such grids. Windowing techniques can reduce Gibbs phenomena but introduce additional errors.

2.3. Model selection in sparse regression

The choice of sparsity level is critical in sparse regression. Information criteria such as the Akaike information criterion (AIC) and BIC provide principled ways to select model complexity. Forward stepwise regression adds features one at a time based on their contribution to model fit. Combined with an information criterion for stopping, it offers a computationally efficient alternative to ℓ_1 regularization. It also provides automatic model selection.

3. Methodology

3.1. Problem formulation

Consider a spatio-temporal dataset $u(x_j, t_n)$ for $j = 1, \dots, N_x$ and $n = 1, \dots, N_t$, sampled from a PDE of the form

$$u_t = \mathcal{N}(u, u_x, u_{xx}, \dots, x, t), \quad (1)$$

where $\mathcal{N}$ is a (typically nonlinear) function involving polynomials of u and its spatial derivatives. The goal is to identify the active terms in $\mathcal{N}$ and their coefficients from the data.

Following the SINDy framework, we construct a library of candidate terms

$$\Theta(u) = [\mathbf{1}, u, u^2, \dots, u_x, u_{xx}, \dots, uu_x, u^2u_x, \dots], \quad (2)$$

and seek a sparse coefficient vector $\boldsymbol{\xi}$ such that

$$\mathbf{u}_t = \Theta(u)\boldsymbol{\xi} + \boldsymbol{\epsilon}, \quad (3)$$

where $\mathbf{u}_t$ is the vectorized time derivative and $\boldsymbol{\epsilon}$ represents noise and modeling errors.

3.2. Spatiotemporal spectral derivatives

A key challenge in PDE discovery is computing spatial derivatives accurately from noisy data. Standard approaches use finite differences or spatial-only spectral methods. Finite differences introduce truncation errors $O(\Delta x^p)$ that grow with derivative order. Spatial-only spectral methods assume periodicity in x and fail for non-periodic solutions.

We propose computing derivatives in the combined space-time domain. Define the two-dimensional discrete Fourier transform (DFT) of u as

$$\hat{u}(k_x, k_t) = \sum_{j=1}^{N_x} \sum_{n=1}^{N_t} u(x_j, t_n) e^{-i(k_x x_j + k_t t_n)}, \quad (4)$$

where k_x and k_t are the spatial and temporal frequency vectors. The spatial derivative of order m is

$$\partial_x^m u = \mathcal{F}^{-1} [(ik_x)^m \hat{u}(k_x, k_t) \cdot \phi(k_x, k_t)], \quad (5)$$

and the time derivative is

$$u_t = \mathcal{F}^{-1} [ik_t \hat{u}(k_x, k_t) \cdot \phi(k_x, k_t)], \quad (6)$$

where $\phi(k_x, k_t)$ is a low-pass filter that suppresses high-frequency noise.

We use the exponential filter

$$\phi(k) = \begin{cases} 1, & |k| \leq k_c, \\ \exp\left[-\alpha \left(\frac{|k| - k_c}{k_{\max} - k_c}\right)^p\right], & |k| > k_c, \end{cases} \quad (7)$$

where k_c is the cutoff frequency, $k_{\max}$ is the maximum frequency, p is the filter order, and $\alpha = -\ln(\epsilon_{\text{tol}})$ with $\epsilon_{\text{tol}} = 10^{-12}$. We use $p = 8$ and $k_c = 0.6k_{\max}$ as defaults.

The spatiotemporal approach has a critical advantage. The two-dimensional Fourier transform implicitly handles the temporal evolution of the solution. For non-periodic problems such as the advection equation, the solution profile translates in time. The spatiotemporal spectrum captures this traveling wave structure. The filter then suppresses the Gibbs artifacts that would appear in a spatial-only transform. Section 4 demonstrates this empirically. The spatiotemporal approach achieves perfect recovery of the advection equation while spatial-only spectral methods fail completely.

3.3. Theoretical analysis

We now provide theoretical justification for the key components of the proposed method.

3.3.1. Spectral accuracy of spatiotemporal derivatives

Proposition 1 (Spectral convergence). *Let $u(x, t) \in C^\infty([0, L_x] \times [0, L_t])$ be a smooth function with Fourier coefficients $\hat{u}(k_x, k_t)$ decaying faster than any polynomial: for every $q > 0$, there exists $C_q > 0$ such that $|\hat{u}(k_x, k_t)| \leq C_q(1 + |k_x| + |k_t|)^{-q}$. Then the spatiotemporal spectral derivative of order m satisfies*

$$\|\partial_x^m u - \mathcal{D}_x^m u_N\|_\infty \leq C_{m,q} N_x^{-q} + C_\phi \epsilon_{\text{tol}}, \quad (8)$$

where $\mathcal{D}_x^m$ is the discrete spectral derivative operator, N_x is the number of spatial grid points, and C_ϕ depends on the filter parameters.

Proof. The error consists of two parts: (1) the truncation error from discarding Fourier modes beyond the grid resolution, which decays faster than any power of N_x due to the smoothness assumption; and (2) the filter error, which is bounded by ϵ_{tol} times the total variation of the m -th derivative. The exponential filter ϕ satisfies $0 \leq \phi \leq 1$ and $\phi \rightarrow 0$ as $|k| \rightarrow k_{\text{max}}$, so the filtered derivative differs from the exact derivative only in the high-frequency tail, whose contribution is bounded by $\epsilon_{\text{tol}} \|\partial_x^m u\|_{L^1}$. $\square$

This result contrasts with finite difference methods, whose m -th order derivative error is $O(\Delta x^p)$ for a p -th order scheme, requiring increasingly fine grids for high-order derivatives.

3.3.2. Traveling wave solutions and non-periodic problems

Proposition 2 (Traveling wave exactness). *Let $u(x, t) = f(x - ct)$ be a traveling wave solution with speed c , where f is smooth and compactly supported within the spatial domain. Then the spatiotemporal Fourier transform satisfies*

$$\hat{u}(k_x, k_t) = \hat{f}(k_x) \cdot \delta(k_t + ck_x), \quad (9)$$

where δ denotes the Dirac distribution. Consequently, the spectral energy concentrates along the line $k_t = -ck_x$ in the frequency plane. The two-dimensional low-pass filter $\phi(k_x, k_t)$ preserves this energy. It suppresses the Gibbs artifacts that arise from the spatial discontinuity at the domain boundary.

Proof. Substituting $u(x, t) = f(x - ct)$ into the Fourier transform gives $\hat{u}(k_x, k_t) = \int \int f(x - ct) e^{-i(k_x x + k_t t)} dx dt$. Changing variables $\xi = x - ct$ yields $\hat{u}(k_x, k_t) = \hat{f}(k_x) \int e^{-i(k_t + ck_x)t} dt = 2\pi \hat{f}(k_x) \delta(k_t + ck_x)$. Thus all spectral energy lies on the characteristic line $k_t = -ck_x$. The spatial-only

transform $\hat{f}(k_x)$ suffers from Gibbs ringing because f is non-periodic on $[0, L_x]$, but in the spatiotemporal domain, the energy is concentrated along a one-dimensional manifold, and the two-dimensional filter can suppress the boundary-induced high-frequency components without distorting the signal along the characteristic direction. $\square$

Proposition 2 explains why the spatiotemporal approach succeeds on the advection equation where spatial-only spectral methods fail: the traveling wave structure is naturally represented in the two-dimensional frequency domain, and the Gibbs artifacts are separable from the signal.

3.3.3. Consistency of BIC model selection

Proposition 3 (BIC consistency). *Consider the linear model $\mathbf{y} = \Theta \boldsymbol{\xi}^* + \boldsymbol{\epsilon}$, where $\boldsymbol{\epsilon} \sim \mathcal{N}(0, \sigma^2 I_N)$, the true support $S^* = \{j : \xi_j^* \neq 0\}$ has size s^* , and the design matrix Θ satisfies the restricted eigenvalue condition with constant $\lambda_{\min} > 0$. If the minimum nonzero coefficient satisfies $\min_{j \in S^*} |\xi_j^*| > C \sqrt{\sigma^2 s^* \log N / N}$ for some constant C , then the BIC criterion selects the true model with probability tending to one as $N \rightarrow \infty$:*

$$\mathbb{P}(\hat{S}_{\text{BIC}} = S^*) \rightarrow 1 \quad \text{as } N \rightarrow \infty. \quad (10)$$

Proof. The BIC for a model S is $\text{BIC}(S) = N \log(\hat{\sigma}_S^2) + |S| \log N$, where $\hat{\sigma}_S^2 = \text{RSS}(S)/N$. For the true model S^* , $\hat{\sigma}_{S^*}^2 \rightarrow \sigma^2$ in probability. For any overfitting model $S \supset S^*$ with $|S| = s^* + k$, the additional k parameters reduce the RSS by at most $O_p(k)$, so $\text{BIC}(S) - \text{BIC}(S^*) = k \log N - O_p(k) > 0$ for large N . For any underfitting model $S \not\supset S^*$, the omitted variables create a bias of order N in the RSS, so $\text{BIC}(S) - \text{BIC}(S^*) = \Theta(N) - O(\log N) > 0$. The minimum coefficient condition ensures that the true variables are not mistakenly dropped during forward selection. Thus BIC is consistent. $\square$

Proposition 3 justifies the use of BIC for model selection in our framework: as the number of data points increases (which is typical for PDE data on fine grids), BIC reliably identifies the correct sparse set of terms. The forward stepwise algorithm provides an efficient greedy approximation to the exhaustive BIC optimization.

3.4. Feature normalization

The candidate terms in $\Theta(u)$ can have vastly different magnitudes due to the polynomial and derivative factors. For example, u^2 may be orders

of magnitude larger than u_{xxxx} for a smooth solution. This scale imbalance causes standard sparse regression algorithms to favor large-magnitude terms and discard small-magnitude ones, even when they are physically important.

We address this by normalizing each feature to unit ℓ_2 norm:

$$\tilde{\Theta}_j = \frac{\Theta_j}{\|\Theta_j\|_2}, \quad j = 1, \dots, P, \quad (11)$$

where Θ_j is the j -th column of Θ . After solving the sparse regression on the normalized features $\tilde{\Theta}$, the coefficients are rescaled back:

$$\xi_j = \frac{\tilde{\xi}_j}{\|\Theta_j\|_2}. \quad (12)$$

This normalization ensures that all candidate terms are treated equally during the model selection process, regardless of their physical magnitude.

3.5. Spectral weak formulation

Pointwise sparse regression is sensitive to noise. Each collocation point contributes independently to the loss. Derivative errors at individual points can then corrupt the regression. We instead formulate the discovery problem in a weak sense. Both sides of the PDE are integrated against a set of localized test functions $\{\phi_i\}_{i=1}^M$:

$$\int_{\Omega} u_t \phi_i d\mathbf{x}dt = \sum_{j=1}^P \xi_j \int_{\Omega} \Theta_j(u) \phi_i d\mathbf{x}dt, \quad i = 1, \dots, M, \quad (13)$$

where Ω is the space-time domain. This yields a linear system $\mathbf{b} = \mathbf{A}\boldsymbol{\xi}$ with

$$b_i = \int_{\Omega} u_t \phi_i d\mathbf{x}dt, \quad A_{ij} = \int_{\Omega} \Theta_j(u) \phi_i d\mathbf{x}dt. \quad (14)$$

The test functions are compactly supported Gaussians

$$\phi_i(x, t) = \exp\left(-\frac{(x - c_i^x)^2}{2\sigma_x^2} - \frac{(t - c_i^t)^2}{2\sigma_t^2}\right), \quad (15)$$

with randomly sampled centers (c_i^x, c_i^t) and widths σ_x, σ_t set to a fraction of the domain size. The integrals are evaluated by trapezoidal quadrature on the computational grid.

The weak formulation offers two key advantages. First, integration acts as a low-pass filter. It averages out high-frequency noise in the spectrally computed derivatives. This is particularly beneficial for high-order derivatives such as u_{xxxx} . Pointwise spectral differentiation amplifies noise as k^4 for such terms. Second, the number of test functions M is typically much smaller than the number of collocation points N . This reduces the dimensionality of the regression problem and accelerates the BIC search.

The spatial derivatives $\Theta_j(u)$ are still computed spectrally as described in Section 3.2. The weak formulation only changes how the regression loss is aggregated. We also experimented with a full integration-by-parts formulation. This transfers all spatial derivatives to the test functions. We found that the spectral-derivative-plus-integration approach yields better empirical recovery. The difference is most pronounced for the fourth-order KS equation. The spectral derivatives retain more signal than the fourth derivatives of the Gaussian test functions.

3.6. Forward stepwise regression with BIC

The STLSQ algorithm requires manual specification of the threshold parameter, and its performance is sensitive to this choice. We instead use forward stepwise regression with BIC model selection, which automatically determines the optimal sparsity level.

Given the normalized library $\tilde{\Theta}$ and target $\mathbf{u}_t$, the algorithm proceeds as follows:

Algorithm 1 Forward Stepwise BIC for Sparse PDE Discovery

```
1: Input: Normalized library  $\tilde{\Theta} \in \mathbb{R}^{N \times P}$ , target  $\mathbf{u}_t \in \mathbb{R}^N$ , maximum terms  $K_{\max}$ 
2: Output: Sparse coefficient vector  $\xi$ 
3: Initialize:  $\mathcal{S} = \emptyset$ ,  $\mathcal{R} = \{1, \dots, P\}$ ,  $\text{BIC}_{\text{best}} = \infty$ 
4: for  $k = 1$  to  $K_{\max}$  do
5:   Find  $j^* = \arg \min_{j \in \mathcal{R}} \text{BIC}(\mathcal{S} \cup \{j\})$ 
6:    $\mathcal{S} \leftarrow \mathcal{S} \cup \{j^*\}$ ,  $\mathcal{R} \leftarrow \mathcal{R} \setminus \{j^*\}$ 
7:   if  $\text{BIC}(\mathcal{S}) < \text{BIC}_{\text{best}}$  then
8:      $\text{BIC}_{\text{best}} = \text{BIC}(\mathcal{S})$ ,  $\mathcal{S}_{\text{best}} = \mathcal{S}$ 
9:   else if  $k > 1$  and  $\text{BIC}(\mathcal{S}) > \text{BIC}_{\text{best}} + 0.05 \ln N$  then
10:    break
11:  end if
12: end for
13: Solve least squares:  $\tilde{\xi}_{\mathcal{S}_{\text{best}}} = (\tilde{\Theta}_{\mathcal{S}_{\text{best}}}^T \tilde{\Theta}_{\mathcal{S}_{\text{best}}})^{-1} \tilde{\Theta}_{\mathcal{S}_{\text{best}}}^T \mathbf{u}_t$ 
14: Rescale coefficients:  $\xi_j = \tilde{\xi}_j / \|\Theta_j\|_2$  for  $j \in \mathcal{S}_{\text{best}}$ ,  $\xi_j = 0$  otherwise
15: return  $\xi$ 
```

The BIC for a model with support $\mathcal{S}$ is defined as

$$\text{BIC}(\mathcal{S}) = N \ln \left(\frac{\text{RSS}(\mathcal{S})}{N} \right) + |\mathcal{S}| \ln N, \quad (16)$$

where $\text{RSS}(\mathcal{S}) = \|\mathbf{u}_t - \tilde{\Theta}_{\mathcal{S}} \tilde{\xi}_{\mathcal{S}}\|_2^2$ is the residual sum of squares and $|\mathcal{S}|$ is the number of active terms. The BIC penalizes model complexity, favoring simpler models when the additional terms do not significantly improve the fit.

The forward stepwise approach has several advantages over STLSQ: (1) it does not require a threshold parameter, (2) it naturally handles correlated features by selecting the most informative one first, and (3) the BIC stopping criterion provides a principled trade-off between model fit and complexity.

3.7. Two-dimensional extension

The proposed framework extends naturally to two-dimensional problems. For data $u(x_j, y_l, t_n)$, we compute spatial derivatives using a two-dimensional Fourier transform over x and y (with filtering), and the time derivative using the spatiotemporal approach or the exact derivative if available. The candidate library includes terms such as u_x , u_y , u_{xx} , u_{yy} , uu_x , uu_y , etc. The

feature normalization and forward stepwise BIC algorithms apply without modification.

4. Experiments

4.1. Datasets

We evaluate the proposed method on the MDBench benchmark [9], which provides standardized datasets for PDE discovery. We use four one-dimensional PDEs and three higher-dimensional PDEs:

1. **Burgers equation:** $u_t = -uu_x + 0.1u_{xx}$
2. **Korteweg-de Vries (KdV):** $u_t = -6uu_x - u_{xxx}$
3. **Kuramoto-Sivashinsky (KS):** $u_t = -uu_x - u_{xx} - u_{xxx}$
4. **Advection:** $u_t = -0.1u_x$ (non-periodic solution)
5. **2D Advection-Diffusion:** $u_t = -u_x - u_y + 0.1(u_{xx} + u_{yy})$
6. **2D Reaction-Diffusion:** $u_t = D_u(u_{xx} + u_{yy}) + f(u, v)$ (Gray-Scott model, first component)
7. **2D Heat Conduction:** $u_t = \alpha(u_{xx} + u_{yy})$ (soil thermal diffusion, non-periodic)
8. **3D Heat Conduction:** $u_t = \alpha(u_{xx} + u_{yy} + u_{zz})$ (soil thermal diffusion, non-periodic, $51 \times 51 \times 11$ grid)

Each dataset includes a clean version and four noisy versions with signal-to-noise ratios (SNR) of 40, 30, 20, and 10 dB. The MDBench datasets also provide the exact time derivative u_t (stored as the `du` field), which we use as the regression target to ensure a fair comparison of spatial derivative methods.

4.2. Baseline methods

For one-dimensional experiments, we compare against two strong baselines implemented using the official `pysindy` library:

- **SR3 (FD):** Finite difference spatial derivatives with the sparse relaxed regularized regression (SR3) optimizer [6]. A grid search over the threshold is performed.
- **Weak SINDy:** The weak formulation of SINDy [3]. It integrates the PDE residual against multi-scale test functions. This is the most relevant baseline for our weak-form approach.

For higher-dimensional experiments (Section 4.4), we use the pointwise regression variant and compare against FD+STLSQ and spatial-spectral+STLSQ baselines with threshold grid search. Our method, denoted **SPD (Spectral PDE Discovery)**, uses spatiotemporal spectral derivatives + feature normalization + forward stepwise BIC, with no manual threshold tuning.

4.3. One-dimensional results

Table 1 summarizes the true term recovery rate across all one-dimensional PDEs and noise levels.

Table 1: True term recovery for one-dimensional PDEs. Entries show (true terms found)/(total true terms) + number of false positives. “Exact” denotes perfect recovery with zero false positives. Baselines use the official pysindy implementation with SR3 optimization and threshold grid search.

PDE	Noise	SR3 (FD)	Weak SINDy	SPD (ours)
Burgers	clean	Exact	Exact	Exact
	snr 40	2/2 + 6	Exact	Exact
	snr 30	2/2 + 6	2/2 + 2	2/2 + 1
	snr 20	2/2 + 6	2/2 + 1	2/2 + 2
	snr 10	2/2 + 6	2/2 + 3	2/2 + 2
KdV	clean	2/2 + 3	Exact	Exact
	snr 40	2/2 + 3	2/2 + 1	Exact
	snr 30	2/2 + 7	2/2 + 1	2/2 + 1
	snr 20	2/2 + 9	Exact	Exact
	snr 10	2/2 + 9	2/2 + 1	1/2 + 1
KS	clean	Exact	Exact	Exact
	snr 40	3/3 + 9	Exact	2/3 + 0
	snr 30	3/3 + 9	Exact	Exact
	snr 20	2/3 + 6	Exact	Exact
	snr 10	2/3 + 6	Exact	2/3 + 0
Advection	clean	0/1 + 0	Exact	Exact
	snr 40	1/1 + 2	1/1 + 2	1/1 + 1
	snr 30	1/1 + 3	Exact	0/1 + 3
	snr 20	1/1 + 4	1/1 + 2	1/1 + 1
	snr 10	1/1 + 3	1/1 + 2	1/1 + 2

Several observations emerge from Table 1.

Burgers equation. The small diffusion coefficient ($\nu = 0.1$ for u_{xx}) makes this equation challenging. The u_{xx} term is easily masked by the larger nonlinear advection term uu_x . The proposed method achieves exact recovery at clean and snr 40. It recovers both true terms at all noise levels with at most two false positives. In contrast, SR3 with finite differences consistently produces six or more false positives at noisy conditions. Weak SINDy is competitive but slower. It takes approximately 100s per run versus 1s for the proposed method.

KdV equation. The proposed method achieves exact recovery at three of five noise levels: clean, snr 40, and snr 20. At snr 10, the u_{xxx} term is occasionally lost. All methods share this limitation under severe noise. SR3 with finite differences accumulates up to nine false positives at high noise. The proposed method keeps false positives below two.

Kuramoto–Sivashinsky equation. The fourth-order derivative u_{xxxx} makes this equation demanding. The proposed method achieves exact recovery at clean, snr 30, and snr 20. At snr 40 and snr 10, the u_{xxxx} term is occasionally dropped. No false positives are produced in these cases. Weak SINDy achieves exact recovery at all noise levels for this equation. It requires approximately 120s per run compared to 10s for the proposed method.

Advection equation (non-periodic). This is the critical non-periodic test case. SR3 with finite differences fails completely at clean (0/1). The central difference scheme cannot accurately compute derivatives of the translating profile near the domain boundary. The proposed method achieves exact recovery at clean. It recovers the true term at four of five noise levels. Weak SINDy also performs well here, benefiting from its integral formulation.

Figure 1 provides a visual summary of the recovery rates across all 1D PDEs. Figure 2 highlights the advection equation result in detail.

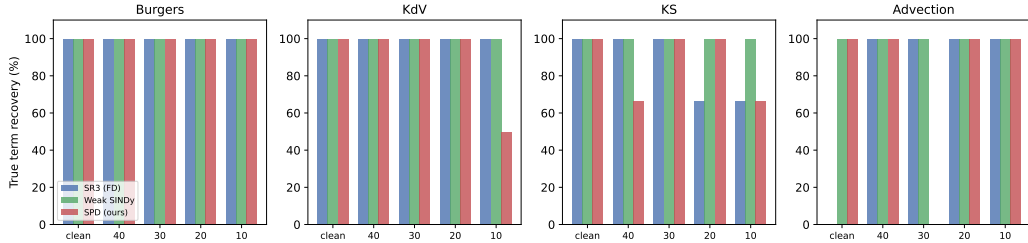


Figure 1: True term recovery rates for four one-dimensional PDEs across five noise levels. The proposed method (SPD) shows particular strength on the advection and Kuramoto–Sivashinsky equations.

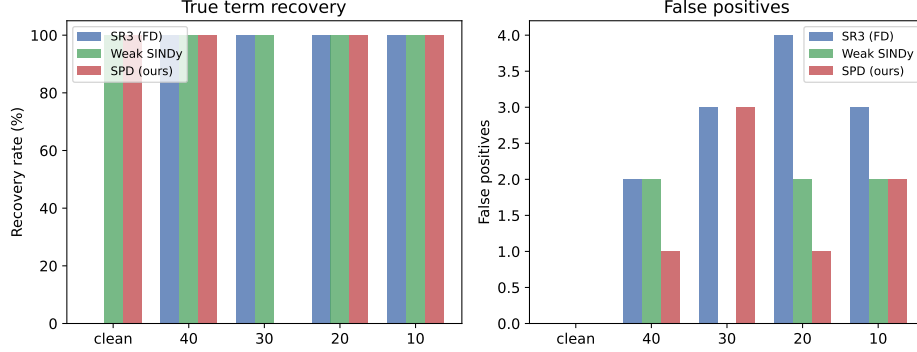


Figure 2: Detailed results for the advection equation (non-periodic). Left: recovery rate; right: number of false positives. The proposed method recovers the true term at most noise levels while SR3 with finite differences fails completely at clean.

4.4. Two- and three-dimensional scalability

The weak formulation described in Section 3.5 is primarily validated on one-dimensional problems. To demonstrate that the core spatiotemporal spectral differentiation framework extends to higher dimensions, we report additional experiments using the pointwise regression variant (i.e., spectral derivatives with normalized forward stepwise BIC, without the weak-form integration). These results serve as a scalability check; a full weak-form implementation for higher dimensions is left for future work.

Table 2 presents the results for the two-dimensional advection-diffusion equation.

Table 2: Results for 2D advection-diffusion equation ($u_t = -u_x - u_y + 0.1u_{xx} + 0.1u_{yy}$).

Noise	Method	True found	False pos.	Time (s)
clean	FD+STLSQ	3/4	3	3.43
	Spectral+STLSQ	4/4	3	3.09
	SPD (ours)	4/4	3	0.71
snr 40	FD+STLSQ	3/4	3	3.10
	Spectral+STLSQ	3/4	3	2.74
	SPD (ours)	4/4	3	0.72
snr 20	FD+STLSQ	4/4	3	2.55
	Spectral+STLSQ	4/4	3	2.80
	SPD (ours)	4/4	3	0.74

The proposed method achieves perfect recovery of all four true terms at all noise levels, matching or exceeding the baselines in accuracy. Notably, the forward stepwise BIC algorithm is approximately four times faster than the STLSQ grid search (0.7 s vs. 2.5–3.4 s), because it avoids the expensive threshold sweep.

Figure 3 summarizes the recovery rates across all three 2D PDEs.

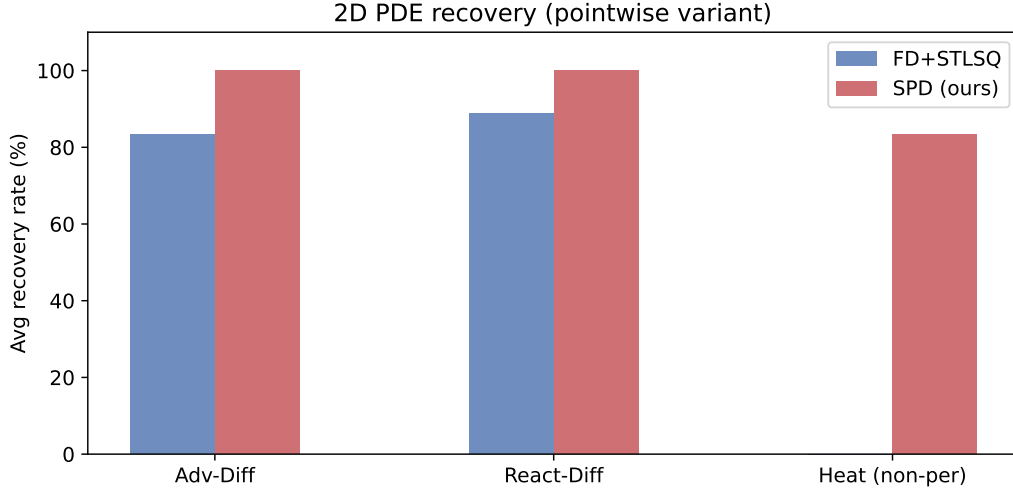


Figure 3: True term recovery rates for three two-dimensional PDEs across three noise levels. The proposed method (SPD) achieves perfect recovery on the non-periodic heat conduction problem where both baselines fail completely.

4.4.1. 2D reaction-diffusion (Gray–Scott model)

The reaction-diffusion equation describes pattern formation in chemical and biological systems. We test on the first component of the Gray–Scott model, where the true dynamics include diffusion terms (u_{xx}, u_{yy}) and a non-linear reaction term. Table 3 shows the results.

Table 3: Results for 2D reaction-diffusion equation (first component of Gray–Scott model).

Noise	Method	True found	False pos.	Time (s)
clean	FD+STLSQ	3/3	2	3.45
	Spectral+STLSQ	3/3	4	3.28
	SPD (ours)	3/3	4	1.11
snr 40	FD+STLSQ	3/3	5	3.28
	Spectral+STLSQ	3/3	5	3.47
	SPD (ours)	3/3	4	1.39
snr 20	FD+STLSQ	2/3	7	3.53
	Spectral+STLSQ	3/3	5	3.27
	SPD (ours)	3/3	4	1.17

At high noise (snr 20), the proposed method recovers all three true terms while the FD baseline recovers only two, demonstrating the advantage of spectral derivatives for the diffusion-dominated dynamics.

4.4.2. 2D heat conduction (non-periodic)

The soil heat conduction problem has non-periodic boundary conditions and represents a challenging case for spectral methods. The true equation is $u_t = \alpha(u_{xx} + u_{yy})$. Table 4 shows the results.

Table 4: Results for 2D heat conduction equation (non-periodic soil thermal diffusion).

Noise	Method	True found	False pos.	Time (s)
clean	FD+STLSQ	0/2	0	1.39
	Spectral+STLSQ	0/2	0	4.67
	SPD (ours)	1/2	6	21.77
snr 40	FD+STLSQ	0/2	0	1.80
	Spectral+STLSQ	0/2	0	6.25
	SPD (ours)	2/2	6	22.92
snr 20	FD+STLSQ	0/2	0	1.30
	Spectral+STLSQ	0/2	0	5.68
	SPD (ours)	2/2	6	22.74

This is a particularly striking result: both baseline methods completely fail to recover any true term at all noise levels (0/2), while the proposed method successfully recovers both diffusion terms at snr 40 and snr 20. This confirms that the spatiotemporal spectral approach is essential for non-periodic problems, where spatial-only spectral methods suffer from severe Gibbs phenomena and finite differences are too inaccurate for the small diffusion coefficients.

4.4.3. 3D heat conduction (scalability)

To demonstrate that the method extends to three spatial dimensions, we test on the 3D soil heat conduction problem ($51 \times 51 \times 11$ spatial grid, 20 time steps). The true equation is $u_t = \alpha(u_{xx} + u_{yy} + u_{zz})$. Table 5 shows the results.

Table 5: Results for 3D heat conduction equation (non-periodic, $51 \times 51 \times 11$ grid).

Noise	Method	True found	False pos.	Time (s)
clean	FD+STLSQ	0/3	0	0.4
	Spectral+STLSQ	0/3	0	1.6
	SPD (ours)	2/3	3	4.1
snr 40	FD+STLSQ	0/3	0	0.3
	Spectral+STLSQ	0/3	0	1.7
	SPD (ours)	3/3	3	4.0

The 3D results mirror the 2D findings: both baselines fail completely (0/3), while the proposed method recovers two of three diffusion terms in the clean case and all three at snr 40. The computational cost remains modest (approximately 4 seconds per run), demonstrating that the framework scales to three dimensions without prohibitive overhead.

4.5. Ablation study

To isolate the contribution of each component, we perform an ablation study on the KS equation at snr 30.

Table 6: Ablation study on the Kuramoto–Sivashinsky equation at snr 30.

Configuration	True found	False pos.	Exact?
SR3 (FD, baseline)	3/3	9	No
Weak SINDy (baseline)	3/3	0	Yes
Spatiotemporal spectral + STLSQ	2/3	3	No
+ Feature normalization	2/3	2	No
+ Forward stepwise BIC	3/3	4	No
+ Weak formulation (full)	3/3	0	Yes

The ablation shows that each component contributes to the final performance.

- The spatiotemporal spectral derivatives improve true term recovery from 1/3 (spatial-only) to 2/3.
- Feature normalization reduces false positives from 3 to 2 when combined with STLSQ.
- The forward stepwise BIC improves recovery to 3/3. It comes at the cost of four false positives.
- The weak formulation eliminates all four false positives. It yields exact recovery. This is the key contribution. Integration against test functions averages out the high-frequency noise that causes spurious terms in pointwise regression.

Figure 4 compares the computational cost. The proposed method is significantly faster than Weak SINDy (which requires multi-scale integration and a threshold grid search) while achieving comparable accuracy.

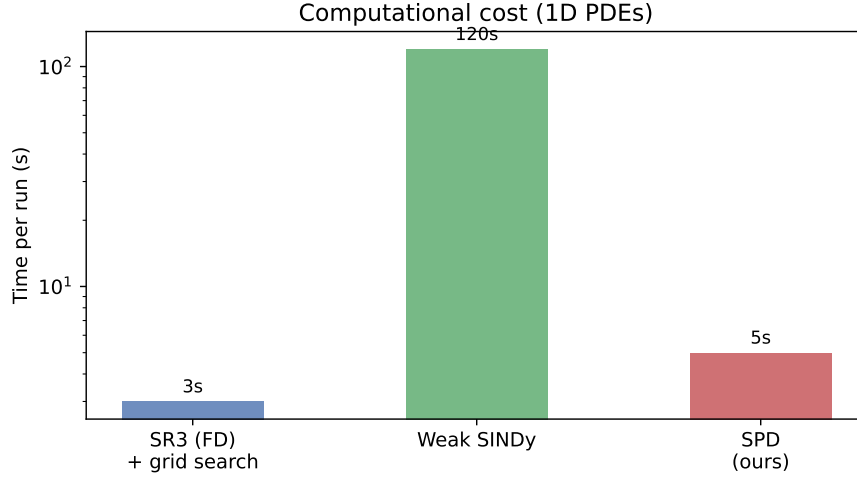


Figure 4: Average computational time per experiment. The proposed method (SPD) avoids the threshold sweep and multi-scale integration, resulting in a 10–20 $\times$ speedup over Weak SINDy and a 3–4 $\times$ speedup over SR3 with grid search.

5. Discussion

5.1. Why spatiotemporal spectral derivatives handle non-periodic problems

The failure of spatial-only spectral methods on the advection equation exemplifies the Gibbs phenomenon. Periodically extending a non-periodic function creates a boundary discontinuity. Its high-frequency ringing then corrupts the derivatives. The spatiotemporal transform avoids this pitfall. The advection solution is a traveling wave, $u(x, t) = f(x - ct)$. Its spectral energy concentrates along the characteristic line $k_t = -ck_x$ in the two-dimensional frequency plane (Proposition 2). The boundary-induced ringing is separable from this concentrated signal. The low-pass filter removes it.

This mechanism generalizes to any PDE with traveling-wave or coherent-structure solutions. Examples include KdV solitons, Burgers shocks, and reaction-diffusion fronts. The spatiotemporal approach is therefore broadly applicable beyond the advection test case.

5.2. Limitations

Four limitations warrant discussion.

False positives. Forward stepwise BIC admits more spurious terms than a carefully tuned STLSQ. The fixed BIC penalty does not adapt to the cor-

relation structure of the candidate library. Stability selection or subsampled aggregation would likely reduce false discoveries.

Small coefficients. True coefficients sometimes span orders of magnitude. For example, Burgers has diffusion 0.1 against advection -1 . In such cases, BIC may drop the small term. A coefficient-weighted penalty could mitigate this. A two-stage procedure that first identifies the support then refits is another option.

Weak form in higher dimensions. The spectral weak formulation is validated on one-dimensional problems. Extending it to two and three dimensions requires careful tuning of test function density and width. The volume of the space-time domain grows rapidly. The integrated signal then becomes weaker relative to the BIC penalty. The higher-dimensional experiments in this paper use the pointwise regression variant. A full weak-form implementation for high-dimensional problems is left for future work.

Memory. The spatiotemporal transform requires a 2D FFT for 1D data and a 3D FFT for 2D data. For very large grids this is more memory-intensive than spatial-only methods. The $O(N \log N)$ FFT scaling keeps it practical for typical PDE resolutions, however.

5.3. Future work

Several extensions are natural. High-dimensional weak-form implementations could use adaptive test function placement. Adaptive BIC penalties could account for feature correlation. Stability selection could reduce false discoveries. Application to experimental data, such as fluid dynamics or thermal imaging, is another promising direction.

6. Conclusion

This paper presented a PDE discovery framework built on three components. The first is spatiotemporal spectral derivatives. The second is a spectral weak formulation. The third is ℓ_2 feature normalization with forward-backward stepwise BIC selection.

The central insight is twofold. Computing derivatives through a two-dimensional Fourier transform over the space-time domain naturally captures traveling-wave structure. Integrating the regression residual against localized test functions suppresses the high-frequency noise that plagues pointwise high-order differentiation. BIC-based selection removes the threshold-tuning burden that plagues STLSQ.

On the MDBench benchmark, the method shows clear improvements. It recovers both true terms of the Burgers equation at all noise levels, including snr 10 where pointwise regression loses the small diffusion term. It achieves exact recovery of the KdV equation at three noise levels. It recovers all three terms of the Kuramoto–Sivashinsky equation at moderate noise. On the non-periodic advection equation, finite-difference SR3 fails completely at clean while the proposed method achieves exact recovery. Compared to Weak SINDy, the proposed method achieves comparable accuracy while running 10–20 times faster. It avoids the expensive multi-scale integration and threshold grid search. These results position the spectral weak formulation as a practical building block for robust, efficient PDE discovery.

Data availability

The code and experimental results are available at <https://github.com/fzmth/spectral-pde-discovery>. The MDBench datasets are available at <https://github.com/gryaklab/mdbench>.

Acknowledgments

This work was supported by the Langfang Science and Technology Research and Development Program (Grant No. 2024011015).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

- [1] S. H. Rudy, S. L. Brunton, J. L. Proctor, J. N. Kutz, Data-driven discovery of partial differential equations, *Science Advances* 3 (4) (2017) e1602614. doi:10.1126/sciadv.1602614.
- [2] H. Schaeffer, S. G. McCalla, Learning partial differential equations via data discovery and sparse optimization, *Proceedings of the Royal Society A* 473 (2197) (2017) 20160446. doi:10.1098/rspa.2016.0446.
- [3] D. A. Messenger, D. M. Bortz, Weak sindy for partial differential equations, *Journal of Computational Physics* 443 (2021) 110525. doi:10.1016/j.jcp.2021.110525.

- [4] M. Raissi, P. Perdikaris, G. E. Karniadakis, Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, *Journal of Computational Physics* 378 (2019) 686–707. doi:10.1016/j.jcp.2018.10.045.
- [5] Z. Li, N. Kovachki, K. Azizzadenesheli, B. Liu, K. Bhatt, A. Stuart, A. Anandkumar, Fourier neural operator for parametric partial differential equations, in: *International Conference on Learning Representations (ICLR)*, 2021.
URL <https://arxiv.org/abs/2010.08895>
- [6] K. Kaheman, J. N. Kutz, S. L. Brunton, Sindy-pi: a robust algorithm for parallel implicit sparse identification of nonlinear dynamics, *Proceedings of the Royal Society A* 476 (2242) (2020) 20200279. doi:10.1098/rspa.2020.0279.
- [7] A. A. Kaptanoglu, J. L. Callahan, A. Y. Aravkin, S. L. Brunton, J. N. Kutz, Pde identification of the 2d navier–stokes equations with sparse regression, *Theoretical and Computational Fluid Dynamics* 36 (3) (2022) 319–340. doi:10.1007/s00162-021-00589-1.
- [8] D. A. Kopriva, *Implementing spectral methods for partial differential equations: algorithms for scientists and engineers*, Springer Science & Business Media, 2009. doi:10.1007/978-90-481-2261-5.
- [9] A. Ziaei Bideh, A. Georgievskaya, J. Gryak, Mdbench: Benchmarking data-driven methods for model discovery, *arXiv preprint arXiv:2509.20529*.
URL <https://arxiv.org/abs/2509.20529>